Hunter Ellis elliswhunter@gmail.com hjkellis.github.io github.com/hjkellis (703)-953-6963

Synopsis

I am an Autonomy Engineer with experience developing software for motion planning, estimation, and control systems.

Experience

Avionics Engineer – Georgia Tech Research Institute
Aug 2025 to Present – Atlanta, GA

- Developing C++ software for collaborative autonomous UAV swarms in a component-based software engineering environment.
- Developed a containerized motion planning service with hybrid planning using graph, sampling, and optimization methods.
- Integrating, developing, and benchmarking sampling-based motion planning algorithms over constraint manifolds – leveraging The Open Motion Planning Library (OMPL).
- Created a world model service for a collaborative-decentralized swarm.
- Wrote data association algorithms for multi-object tracking (MOT) and vehicle-to-vehicle track consensus.
- Software systems integration

Robotics Researcher / M.S. Thesis – Virginia Tech
Aug 2023 to May 2025 – Blacksburg, VA

- Built a 6-DOF robotic arm and an accompanying ROS2–Gazebo simulation environment for training, evaluating, and deploying custom control algorithms.
- Integrated object detection (YOLOv8), natural language processing, symbolic reasoning, and a DDPG-based reinforcement learning policy for robotic manipulation tasks.
- Aided professors in teaching fundamental concepts in linear systems theory and digital signal processing, including Laplace Transforms, Z-Transforms, system stability, and FIR & IIR filter design.
- Assisted with hands-on projects to illustrate and integrate analog and digital filter design and application on breadboards and TI MSP432 development boards.

Thrust Vector Control Intern – Jacobs Space Exploration Group
May 2024 to Aug 2024 – Huntsville, AL

- Developed thrust vector control testing hardware and software for NASA's Active Inertial Load Simulator at the Marshall Space Flight Center.
- Created and ran tests to develop a mathematical model of an electro-mechanical actuator – used Python, MATLAB, and LabVIEW.

Control Research Intern – Grenoble Electrical Engineering Lab
Jun 2023 to Aug 2023 – Grenoble, FR

- Implemented inverter control methods for 4-leg (microgrid) topologies and tested them in Simulink and HIL.

Engineering Intern (NREIP) – Naval Surface Warfare Center
Jun 2022 to Aug 2022 – Bethesda, MD

- Developed concept hospital sea-train designs, estimated fuel consumption, and electrical power loads.

Education

M.S. Computer Engineering
Virginia Tech May 2025

- Focus: Software and Machine Intelligence
- Research Group: COOL Autonomy Lab @ UT Austin

B.S. Computer Engineering
B.S. Electrical Engineering
Virginia Tech May 2024

- Focus: Controls, Robotics, and Autonomy

Tools/Libs

ROS2, OMPL, DDS, GNU/Linux, Git, Qt, Simulink, Docker, FreeRTOS, SOLIDWORKS, Autodesk Inventor